

AcademicProposal – Content Overview

Status: ✓

Version: 0.2.1

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The *AcademicProposal* folder contains all scientific and institutional documents describing the C-ABI-Bridge-AOT technology in an academic, research-oriented, and security-critical context. The structure is designed to give readers a clear and immediate overview of motivation, scientific relevance, research directions, and sector-specific impact.

FH / Bachelor / Applied Sciences Relevance

The C-ABI-Bridge-AOT technology is well suited for applied sciences programs, including FH (Universities of Applied Sciences), Bachelor-level work, and HF technical diplomas. Students can focus on practical implementations, interoperability testing, security evaluations, or reproducible build pipelines. The architecture provides a clear technical foundation that is both practical and academically robust.

1. [ExecutiveSummary.md](#)

Purpose:

Concise, institution-ready summary of the entire project.

Suitable for professors, research groups, government agencies, financial institutions, military organizations, and industry stakeholders.

Contents:

- Problem Statement
 - Concept Overview
 - Scientific Relevance
 - Research Objectives
 - Expected Outcomes
 - Institutional Impact
 - Conclusion & Recommendation
-

2. [AcademicProposalDE.md](#)

Purpose:

Full scientific description in German.

Contents:

- Introduction
 - Motivation
 - Scientific Research Areas
 - Master's Thesis Proposal
 - PhD Dissertation Proposal
 - Academic Level Distinction (Bachelor/HF/Master/PhD)
 - Sector Relevance (Government, Finance, Military, Public Infrastructure)
 - Invitation to Collaborate
 - Conclusion / Vision
-

3. [AcademicProposalEN.md](#)

Purpose:

International version for universities, institutes, government bodies, and industry.

Contents:

- Introduction
 - Motivation
 - Scientific Research Areas
 - Master's Thesis Proposal
 - PhD Dissertation Proposal
 - Academic Level Distinction
 - Sector Relevance
 - Invitation to Collaborate
 - Conclusion / Vision
-

4. [ResearchRoadmap Master.md](#)

Purpose:

Structured research roadmap for Master's level work.

Contents:

- Research Objectives

- Research Questions
 - Scientific Domains
 - Methodology
 - Milestones
 - Timeline
 - Resources
 - Risks & Challenges
 - Expected Outcomes
 - Long-Term Vision
-

5. [ResearchRoadmap PhD.md](#)

Purpose:

Structured research roadmap for PhD-level work.

Contents:

- Formal Research Objectives
 - Formal Research Questions
 - Scientific Domains
 - Formal Methods (Coq, Dafny, Lean, TLA+, Z3)
 - Milestones
 - Timeline (3+ years)
 - Resources
 - Risks & Challenges
 - Expected Outcomes
 - Long-Term Vision
-

6. [Roadmap ASCII.md](#) and [Roadmap-C-ABI-Bridge-Aot](#)

Purpose:

Visual ASCII representation of the entire roadmap.

Useful for README files, presentations, and quick orientation.

Contents:

- [Master Roadmap \(ASCII\)](#)
- [PhD Roadmap \(ASCII\)](#)
- Master vs. PhD comparison
- Compact milestone overview

7. Optional Security-Attacks Context

While the C-ABI-Bridge-AOT Academic Proposal focuses primarily on architecture, interoperability, reproducibility, and cryptographic module design, it is important to acknowledge that security-attack analysis represents a significant and independent research domain. A full exploration of attack vectors, threat models, and adversarial strategies can easily constitute an entire thesis or diploma project on its own. For this reason, security-attacks are referenced as an optional extension rather than a core component of this document.

Students or researchers who wish to pursue deeper security analysis may integrate attack-focused methodologies, formal threat modeling, or cryptographic vulnerability studies as part of their thesis work.

8. Additional Optional Files

These files can be added later to extend the AcademicProposal folder:

- **README_Short.md** – Short summary for repository entry
 - **ArchitectureDiagram_ASCII.md** – ASCII diagram of the C-ABI-Bridge architecture
 - **NormativeMapping.md** – Mapping of FIPS/NIST/BSI requirements
 - **FormalModelingNotes.md** – Notes for formal verification work
 - **PublicationPlan.md** – Plan for academic publications
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9. ETH/EPFL Academic Edition

For ETH/EPFL students and researchers, a dedicated academic version of the proposal is available. It highlights the scientific motivation, formal modeling options, deterministic execution, normative context, and advanced Master/PhD research directions relevant to high-level academic work.

See: [AcademicProposal ETH EPFL EN.md](#)

Summary

The *AcademicProposal* folder provides the complete scientific foundation for the development and evaluation of the C-ABI-Bridge-AOT technology.

It is clearly structured, institution-ready, and covers all relevant areas: architecture, scientific analysis, normative standards, research planning, security considerations, and societal impact.

Prepared by:

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Architect & Engineering – Cryptographie, NativeAOT / C-ABI Interoperability
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Call for Students – Write Your Thesis with C-ABI-Bridge-AOT

Are you looking for an exciting, practical, and academically solid topic for your Bachelor, FH, HF, or Master thesis?

The C-ABI-Bridge-AOT technology provides an ideal foundation.

It combines modern cryptography, interoperability, compiler behavior, reproducible builds, and security-critical architecture in a well-structured project that is both technically challenging and academically valuable.

If you are interested in secure modules, C-ABI interoperability, NativeAOT, deterministic execution, or reproducible cryptography, you will find concrete research questions, roadmaps, and guidance for your thesis here.

You can propose this topic directly to your professor – the [Executive Summary](#) helps you present the scientific relevance clearly and convincingly.

Call for Students – Thesis Opportunities with C-ABI-Bridge-AOT

Why this topic?

The C-ABI-Bridge-AOT technology offers a unique combination of practical software engineering and academic depth. It is an excellent choice for students seeking a modern, relevant, and technically challenging topic for their thesis work.

Who is this for?

- Bachelor students
- FH students (Universities of Applied Sciences)
- HF students (Higher Technical Colleges)
- Master students
- Anyone interested in cryptography, security, interoperability, compiler design, software engineering, or system architecture

What can you explore?

- Interoperability between NativeAOT applications and native C modules
- Security analysis of ABI boundaries
- Reproducible build pipelines and deterministic execution
- Cryptographic module architecture and secure interface design
- Normative requirements (FIPS, NIST, BSI)
- Compiler behavior and cross-language integration
- Formal modeling and verification (for advanced Master or PhD work)

Why is this valuable?

- Practical and industry-relevant
- Scientifically grounded
- Clearly structured research directions
- Suitable for Bachelor, FH, HF, and Master theses
- Relevant for industry, government, finance, defense, and research institutions
- Supported by a full Academic Proposal and Executive Summary

How to get started

1. Read the [Executive Summary](#)
2. Explore the Academic Proposal ([EN/DE](#)) or ETH/EPFL ([EN/DE](#))
3. Choose a topic from the [Master](#) or [PhD](#) research roadmaps
4. Present the topic to your professor or supervisor
5. Begin your thesis with a clear scientific foundation and well-defined goals

Available Documents

- [ExecutiveSummary.md](#) – institutional overview
 - [ResearchRoadmap-PhDLevel-EN.md](#) – PhD-level-Roadmap
 - [ResearchRoadmap-MasterLevel-EN.md](#) – Master-level-Roadmap
 - [AcademicProposalDE.md](#) / [AcademicProposalEN.md](#) – full scientific description
 - [AcademicProposal ETH EPFL DE.md](#) / [AcademicProposal ETH EPFL EN.md](#) - scientific description for ETH / EPFL
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Why professors appreciate this topic

Although professors are often busy, this topic is easy to accept because:

- it is clearly documented,
- it has strong scientific relevance,
- it is modern and technically meaningful,
- it requires no additional preparation from them,
- students can present it independently using the Executive Summary.

This makes C-ABI-Bridge-AOT an ideal candidate for student-driven thesis proposals.

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Executive Summary – C-ABI-Bridge-AOT Technology

Status: ✓

Version: 0.2.1

Last Updated: 2026.07.16

General Institutional Version (Universities, Research Institutes, Government, Finance, Military, Industry)

1. Problem Statement

Modern cryptographic systems face increasing pressure to deliver **security, reproducibility, interoperability**, and **long-term auditability**.

Critical sectors — including government agencies, financial institutions, military systems, and public digital infrastructure — require cryptographic modules that:

- operate deterministically
- expose minimal and stable interfaces
- are reproducible across compilers and platforms
- can be audited and certified over long lifecycles
- integrate safely across multiple programming languages

Existing solutions often suffer from:

complex runtime dependencies

- non-deterministic build pipelines
- language-specific bindings with unclear boundaries
- difficult certification processes
- limited interoperability

A new architectural approach is needed to address these challenges.

2. Concept Overview

The **C-ABI-Bridge-AOT** technology introduces a **minimalistic, reproducible, language-agnostic Application Binary Interface (ABI)** for cryptographic modules.

Its core principles:

- Clear module boundaries
- Deterministic execution
- Reproducible builds
- Language-independent interoperability

- Minimal attack surface
- Audit-friendly architecture

By separating cryptographic logic from language runtimes, the architecture enables:

- stable cross-language integration (C#, Rust, Go, Python, Java, C++)
- predictable behavior across compilers (Clang, GCC, MSVC)
- simplified certification (FIPS 140-3, NIST SP 800 series)
- long-term maintainability and auditability
- This makes the technology suitable for both academic research and real-world deployment in critical environments.

3. Scientific Relevance

The project intersects with multiple scientific domains:

- Cryptography & Security Architecture
- ABI Design & Interoperability
- Compiler Theory & Deterministic Build Pipelines
- Normative Standards (FIPS, NIST, BSI)
- Formal Verification & Mathematical Modeling
- Supply-Chain Security & Reproducible Systems

It provides a foundation for:

- Bachelor/HF practical implementations
- Master-level scientific evaluations
- PhD-level formal verification and theoretical contributions
- interdisciplinary research collaborations

The architecture addresses a **current research gap**: secure, reproducible, cross-language cryptographic modules with formally definable boundaries.

4. Research Objectives

The overarching objectives include:

- analyzing the architectural design
- evaluating security and module boundaries
- assessing interoperability across languages
- mapping normative requirements (FIPS, NIST)
- modeling deterministic and reproducible execution
- developing formal proofs of safety (PhD level)
- producing scientific documentation and publications
- defining certification-ready module structures

These objectives support both academic research and institutional decision-making.

5. Expected Outcomes

The project is expected to deliver:

- a scientifically validated architectural model
- security and interoperability assessments
- normative compliance analysis
- reproducibility and compiler behavior evaluation
- formal mathematical models (PhD)
- verified security properties (PhD)
- certification readiness roadmap
- recommendations for critical sectors
- academic publications and long-term research value

These outcomes benefit both research communities and operational institutions.

6. Institutional Impact

Universities & Research Institutes

- clear research questions
- interdisciplinary relevance
- strong publication potential
- compatibility with formal verification research
- suitability for Master's and PhD programs

Government Agencies

- audit-friendly architecture
- reproducible cryptographic modules
- long-term maintainability
- simplified certification pathways

Financial Institutions

- regulatory compliance support
- stable and deterministic interfaces
- secure cross-language integration
- reduced operational risk

Military & Defense

- deterministic AOT-compatible cryptography
- minimal attack surface
- reproducible builds for mission-critical systems
- formal verification potential

Industry & Critical Infrastructure

- transparent and maintainable security modules
- reduced supply-chain risk
- interoperability across heterogeneous systems

Public Sector & Society

- trustworthy open-source cryptography
- secure digital services
- transparent architecture for public review

7. Conclusion & Recommendation

The C-ABI-Bridge-AOT technology provides a **strategically valuable, scientifically grounded, and operationally relevant** foundation for the next generation of secure cryptographic modules.

Its minimalistic and reproducible design addresses key challenges faced by academia, government, finance, defense, and public infrastructure.

Institutions are encouraged to:

- support research based on this architecture
- integrate the technology into security evaluations
- explore formal verification and certification pathways
- collaborate across sectors to advance secure cryptographic systems

The project represents a **forward-looking investment** in digital security, scientific innovation, and long-term infrastructure resilience.

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Academic Proposal – Scientific Advancement of the C-ABI-Bridge-AOT Technology

Status: ✓

Version: 0.2.1

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1. Introduction

Modern cryptographic systems face increasing demands for security, reproducibility, interoperability, and long-term auditability. In critical domains such as government agencies, financial institutions, military systems, and public digital infrastructure, cryptographic modules must operate reliably across different platforms and programming languages while maintaining deterministic behavior and clearly defined module boundaries.

The C-ABI-Bridge-AOT technology presented here introduces an innovative architectural approach for providing cryptographic functionality through a minimalistic, reproducible, and language-agnostic C Application Binary Interface (ABI). By separating cryptographic core logic from language-specific runtime environments, the architecture enables stable, auditable, and interoperable modules that can be used across diverse ecosystems.

This document serves as an academic proposal outlining the scientific relevance of the technology and presenting potential research directions. The project is not intended as a finished product but as a foundation for Master's theses, doctoral dissertations, and interdisciplinary research in cryptography, software architecture, compiler theory, interoperability, and normative security standards. At the same time, the technology remains accessible to students and practitioners at all levels: Bachelor, FH and HF students can explore practical implementations, while Master's and PhD candidates can investigate deeper scientific and normative aspects.

By openly publishing the technology and its academic context, this proposal invites collaboration, critical analysis, and further development. Innovation often begins with simple, well-structured ideas — and this project aims to provide exactly such a starting point.

2. Motivation

The motivation for advancing the C-ABI-Bridge-AOT technology arises from several challenges in modern cryptographic engineering. Critical sectors increasingly require cryptographic modules that are secure, reproducible, interoperable, and certifiable. The complexity of contemporary software ecosystems often leads to cryptographic logic being distributed across multiple layers, complicating audits, certification processes, and long-term maintenance.

The proposed architecture addresses these challenges by offering a minimalistic and clearly defined ABI boundary. This separation reduces runtime complexity, improves auditability, and enables deterministic execution — a key requirement for normative standards such as FIPS 140-3. Reproducible builds and deterministic behavior are essential for mitigating supply-chain risks and ensuring trust in cryptographic modules deployed in sensitive environments.

The scientific motivation is further strengthened by the interdisciplinary nature of the technology. It touches on cryptography, compiler theory, software architecture, interoperability, formal

verification, and normative standardization. This makes the project an ideal foundation for academic work at various levels, from practical implementations to theoretical research.

Finally, the open nature of the project encourages collaboration between academia, industry, and public institutions. By providing a transparent and extensible foundation, the technology supports long-term innovation and contributes to the development of secure and trustworthy digital systems.

3. Scientific Research Areas

The C-ABI-Bridge-AOT technology intersects with numerous scientific fields, offering a broad foundation for academic exploration:

3.1 Cryptography and Security Architecture

- Minimal and auditable cryptographic APIs
- Deterministic execution models
- Secure separation of cryptographic core and runtime environment
- Modular security architecture for critical systems

3.2 ABI Design and Interoperability

- Formal modeling of ABI layouts
- Stable and reproducible interfaces
- Cross-language interoperability (C#, Rust, Go, Python, Java, C++)
- Analysis of calling conventions and struct packing
- Attack-surface minimization through deterministic boundaries

3.3 Compiler and Runtime Theory

- Deterministic build pipelines
- Compiler independence (Clang, GCC, MSVC)
- Impact of optimizations on security
- AOT-compatible cryptography for critical environments

3.4 Normative Standards and Certification

- FIPS 140-3 compliance analysis
- NIST SP 800-series requirements
- Formal modeling of module boundaries
- Self-test and DRBG integration

3.5 Formal Verification and Mathematical Modeling

- Formal proofs of memory safety
- Deterministic API semantics
- Verification of interop boundaries
- Use of Coq, Dafny, Lean, TLA+, Z3

3.6 Security Analysis and Threat Modeling

- Interop-related attack vectors
- Memory ownership risks
- Secure module boundary design
- Minimization of dynamic code paths

3.7 Reproducible Systems and Supply-Chain Security

- Reproducible builds
- Deterministic modules
- Audit-friendly pipelines
- Minimalistic design as a defense mechanism

3.8 Open-Source Research and Societal Impact

- Collaborative development
- Transparency in cryptographic systems
- Public trust in digital infrastructure

4. Master's Thesis Proposal

Title

Evaluation and Scientific Analysis of a Reproducible, Language-Agnostic C-ABI-Bridge for Secure Cryptographic Modules

Objectives

- Architectural analysis
- Security evaluation
- Interoperability assessment
- Normative comparison (FIPS 140-3)
- Roadmap for certification readiness

Research Questions

- How can ABI boundaries be designed to be stable and reproducible?
- What security benefits arise from minimalistic API design?
- How does the architecture behave across multiple languages?
- Which FIPS requirements are inherently supported?

Expected Outcomes

- Scientific documentation
- Security and interoperability analysis
- Certification roadmap
- Recommendations for critical sectors

5. PhD Dissertation Proposal

Title

Formal Modeling, Verification, and Normative Analysis of a Reproducible C-ABI-Cryptography Bridge for Critical Systems

Objectives

- Formal mathematical modeling
- Security proofs
- Normative verification
- Compiler-theoretical analysis
- Scientific publications

Research Questions

- How can ABI boundaries be formally modeled?
- How can deterministic execution be proven mathematically?
- Which FIPS requirements can be formally verified?
- How do compiler optimizations affect formal security?

Expected Outcomes

- Formal models
- Verified security properties
- Normative compliance analysis
- Peer-reviewed publications

6. Academic Level Distinction (Bachelor / FH / HF / Master / PhD)

Bachelor / FH / HF

- Practical implementation
- Basic architectural understanding
- Introductory interoperability tests

Master

- Scientific analysis
- Normative evaluation
- Security and interoperability studies

PhD

- Formal modeling
- Mathematical verification
- Theoretical contributions
- Publications

All levels contribute meaningfully — the distinction lies in analytical depth, not personal value.

7. Relevance for Government, Finance, Military, and Society

Government

- Auditability
- Reproducibility
- Deterministic modules

Finance

- Regulatory compliance
- Stable interfaces
- Secure interoperability

Military

- AOT-compatible cryptography
- Minimal attack surface
- Deterministic behavior

- Transparent open-source cryptography
- Trustworthy digital services

8. Invitation to Collaborate

This project welcomes contributions from students, researchers, institutions, and practitioners at all levels. Collaboration strengthens the scientific foundation and expands the practical applicability of the technology.

9. Conclusion / Vision

The C-ABI-Bridge-AOT technology represents a vision for secure, reproducible, and interoperable cryptographic modules. By providing a clear and scientifically grounded foundation, it invites further research, innovation, and collaboration. The long-term goal is to support the development of trustworthy cryptographic systems that serve critical sectors and the broader public.

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Academic Proposal – ETH/EPFL Edition

C-ABI-Bridge-AOT – Architecture, Interoperability & Cryptography

1. Scientific Motivation

The C-ABI-Bridge-AOT technology addresses a central challenge in modern system architecture: reproducible, deterministic, and security-critical interoperability between Ahead-Of-Time compiled modules and native C libraries. This topic is highly relevant in both theoretical and practical research, touching compiler design, cryptography, system security, and formal modeling.

1.1 Reference to the Executive Summary

A concise institutional overview of the project is available in the file [ExecutiveSummary.md](#). It summarizes the key architectural, security-related, and organizational aspects and serves as a high-level entry point for professors and academic decision-makers before engaging with the deeper scientific content of this ETH/EPFL edition.

1.2 Reference to the Main Academic Proposals (DE/EN)

For a complete and general academic description of the C-ABI-Bridge-AOT architecture, the main proposals [AcademicProposalDE.md](#) and [AcademicProposalEN.md](#) are available. These documents provide the foundational architectural, interoperability, and cryptographic context upon which this ETH/EPFL edition builds. They serve as the primary reference for students and researchers before engaging with the advanced scientific extensions presented here.

2. Architectural Foundations

The architecture defines a deterministic and strictly bounded C-ABI interface independent of runtime environments. This enables:

- reproducible builds,
- deterministic execution,
- cross-language interoperability,
- secure cryptographic module boundaries,
- formal analyzability.

The minimalistic structure is intentional to support formal reasoning and scientific analysis.

3. Formal Modeling (optional)

For ETH/EPFL students, the architecture provides an ideal foundation for:

- formal specifications,
- modeling ABI boundaries,
- mathematical descriptions of deterministic modules,
- formal security proofs,

- modeling interoperability under adversarial conditions.

This section is optional and intended for advanced Master or PhD research.

4. Security-Attacks & Threat Modeling (optional)

Security-attack analysis is a standalone research domain.

A full exploration of attack vectors can easily constitute an entire thesis. Therefore, this topic is referenced as an optional extension rather than a core component.

Possible directions:

- ABI boundary attacks,
- side-channel analysis,
- memory-safety vulnerabilities,
- reproducibility manipulation,
- formal threat-modeling methodologies.

5. Deterministic Execution & Reproducibility

The architecture enables:

- deterministic cryptographic operations,
- reproducible build pipelines,
- stable ABI layouts,
- formal traceability.

This is essential for research and high-security systems.

6. Normative Context (FIPS/NIST/BSI)

The architecture aligns with:

- FIPS-140 module boundaries,
- NIST cryptographic guidelines,
- BSI security requirements.

These standards can serve as a foundation for advanced academic work.

7. Research Directions (Master/PhD)

- formal modeling of ABI boundaries,
- mathematical analysis of deterministic modules,
- interoperability security under adversarial conditions,
- reproducible cryptography,
- normative security architecture,

- compiler behavior and AOT optimization,
- cross-language security design.

8. Why ETH/EPFL Might Be Interested

- clear scientific questions,
 - formal modeling potential,
 - high security relevance,
 - modern architectural design,
 - reproducible systems,
 - cryptographic significance,
 - interoperability research potential.
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■ Structured Research Roadmap (Master's Level)

C-ABI-Bridge-AOT Technology

1. Research Objectives

- Evaluate the architectural design of the C-ABI-Bridge-AOT.
- Analyze security properties and module boundaries.
- Assess interoperability across multiple programming languages.
- Compare the architecture against normative standards (FIPS 140-3, NIST SP 800).
- Develop a roadmap for certification readiness.
- Document scientific findings for academic and institutional use.

2. Research Questions

- How can a minimalistic ABI boundary improve auditability and security?
- Which FIPS 140-3 requirements are inherently supported by the architecture?
- How does the bridge behave across languages such as C#, Rust, Go, Python?
- What risks arise from cross-language interoperability?
- How reproducible are builds across compilers and platforms?

3. Research Domains

- Cryptography & Security Architecture
- ABI Design & Interoperability
- Compiler & Runtime Behavior
- Normative Standards (FIPS, NIST)
- Reproducible Systems
- Threat Modeling & Security Analysis

4. Methodology

Technical Methods

- ABI layout inspection
- Interop testing across languages
- Compiler comparison (Clang, GCC, MSVC)
- Build reproducibility evaluation

Normative Methods

- FIPS 140-3 requirement mapping
- NIST SP 800-series analysis
- Audit simulation

Security Methods

- Threat modeling
- Memory ownership analysis
- Boundary attack surface evaluation

5. Milestones

- M1: Architectural documentation completed
- M2: Interop test suite implemented
- M3: Normative analysis completed
- M4: Security evaluation completed
- M5: Certification readiness roadmap drafted
- M6: Final scientific documentation delivered

6. Timeline

Short-Term (0–6 months)

- Architecture analysis
- Interop testing
- Initial normative mapping

Mid-Term (6–12 months)

- Security analysis
- Compiler & reproducibility evaluation

Long-Term (12–18 months)

- Certification roadmap
- Final thesis documentation

7. Resources

- Programming languages: C#, Rust, Go, Python, C++
- Compilers: Clang, GCC, MSVC
- Access to FIPS/NIST documentation
- Open-source repository

8. Risks & Challenges

- Complexity of normative interpretation
- Interop-related security risks

- Compiler inconsistencies
- Limited reproducibility across platforms

9. Expected Outcomes

- Scientific evaluation of the architecture
- Security and interoperability analysis
- Normative compliance assessment
- Certification roadmap
- Recommendations for government, finance, military, public infrastructure

10. Long-Term Vision

- Foundation for secure, reproducible, certifiable cryptographic modules.
 - Contribution to academic research and critical infrastructure security.
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■ Structured Research Roadmap (PhD Level)

Formal Modeling & Verification of the C-ABI-Bridge-AOT Technology

1. Research Objectives

- Develop formal mathematical models of the ABI boundary.
- Prove security properties using formal verification tools.
- Analyze normative requirements (FIPS 140-3, NIST SP 800) at a theoretical level.
- Investigate compiler behavior and deterministic build pipelines.
- Produce peer-reviewed scientific publications.
- Establish theoretical foundations for certifiable cryptographic modules.

2. Research Questions

Formal Questions

- How can ABI boundaries be formally modeled?
- Which mathematical properties guarantee deterministic execution?
- How can memory safety be proven formally?

Normative Questions

- Which FIPS 140-3 requirements can be formally verified?
- How can module boundaries be expressed mathematically?

Compiler-Theoretical Questions

- How do compiler optimizations affect formal security proofs?
- How can reproducible builds be modeled theoretically?

Interoperability Questions

- How can cross-language interoperability be formally verified?
- What mathematical constraints ensure safe interop boundaries?

3. Research Domains

- Formal Verification
- Mathematical Modeling
- Compiler Theory
- Normative Security Standards
- Cryptographic Architecture
- Interoperability Theory

- Reproducible Systems
- Supply-Chain Security

4. Methodology

Formal Methods

- Coq
- Dafny
- Lean
- TLA+
- Z3 Solver

Theoretical Methods

- Mathematical modeling
- Formal semantics
- Deterministic execution proofs

Normative Methods

- FIPS 140-3 formal requirement mapping
- NIST SP 800-series theoretical analysis

5. Milestones

- M1: Formal ABI model defined
- M2: Deterministic execution model proven
- M3: Memory safety proofs completed
- M4: Normative formal mapping completed
- M5: Compiler behavior model established
- M6: Peer-reviewed publications submitted
- M7: Dissertation completed

6. Timeline

Short-Term (0–12 months)

- Formal modeling
- Initial proofs
- Normative theoretical mapping

Mid-Term (12–24 months)

- Extended formal verification
- Compiler-theoretical analysis
- Publication of initial results

Long-Term (24–36+ months)

- Full formal verification
- Dissertation writing
- Publication of final results

7. Resources

- Formal verification tools (Coq, Dafny, Lean, TLA+, Z3)
- Compiler toolchains
- Access to normative standards
- Academic supervision and peer review

8. Risks & Challenges

- High complexity of formal models
- Difficulty proving normative requirements
- Compiler nondeterminism
- Time-intensive verification processes

9. Expected Outcomes

- Formal mathematical models
- Verified security properties
- Normative compliance proofs
- Compiler-theoretical insights
- Peer-reviewed publications
- Dissertation with long-term scientific relevance

10. Long-Term Vision

- Establishment of a formally verified, certifiable cryptographic module architecture.
 - Contribution to international research in cryptography and formal verification.
 - Foundation for secure digital infrastructure in government, finance, military, and public sectors.
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Architect & Engineering – Cryptographie, NativeAOT / C-ABI Interoperability
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C-ABI-BRIDGE-AOT – STRUCTURED RESEARCH ROADMAP		
MASTER LEVEL ROADMAP	PhD LEVEL ROADMAP	
1. OBJECTIVES	1. OBJECTIVES	
- Architecture evaluation	- Formal ABI model	
- Security analysis	- Mathematical verification	
- Interoperability assessment	- Normative formal analysis	
- Normative mapping (FIPS/NIST)	- Compiler/build theory	
- Certification readiness	- Peer-reviewed publications	
2. RESEARCH QUESTIONS	2. RESEARCH QUESTIONS	
- Minimal ABI & auditability	- Formal ABI modeling	
- FIPS coverage	- Deterministic execution	
- Interop risks	- Memory-safety proofs	
- Reproducible builds	- Formal FIPS verification	
	- Interop formal constraints	
3. DOMAINS	3. DOMAINS	
- Cryptography & architecture	- Formal verification	
- ABI design & interop	- Mathematical modeling	
- Compiler behavior	- Compiler theory	
- Normative standards	- Normative security	
- Threat modeling	- Reproducible systems	
4. METHODS	4. METHODS	
- ABI inspection	- Coq, Dafny, Lean, TLA+, Z3	
- Interop tests	- Formal semantics	
- Compiler comparison	- Proof construction	
- Threat modeling	- Theoretical FIPS/NIST mapping	
5. MILESTONES	5. MILESTONES	
M1 Architecture documented	M1 Formal ABI model defined	
M2 Interop suite implemented	M2 Determinism proven	
M3 Normative analysis completed	M3 Memory-safety proofs done	

M4 Security evaluation completed	M4 Normative formal mapping	
M5 Certification roadmap drafted	M5 Compiler behavior model	
M6 Final thesis delivered	M6 Publications submitted	
+-----+		
6. TIMELINE	6. TIMELINE	
0-6m: Architecture, interop	0-12m: Modeling & initial proofs	
6-12m: Security, compiler	12-24m: Extended verification	
12-18m: Certification roadmap	24-36m+: Full proofs, thesis	
+-----+		
7. OUTCOMES	7. OUTCOMES	
- Architecture evaluation	- Formal mathematical models	
- Security & interop analysis	- Verified security properties	
- Normative compliance assessment	- Normative compliance proofs	
- Certification roadmap	- Publications & dissertation	
- Recommendations for critical sectors	- Long-term scientific impact	
+-----+		
8. VISION	8. VISION	
Secure, reproducible, certifiable	Formally verified, globally	
cryptographic modules for critical	certifiable cryptographic	
infrastructure & public digital systems	architecture for critical infrastructure	
+=====+		

+-----+		
Structured Research Roadmap – Overview		
+-----+		
MASTER LEVEL	PhD LEVEL	
+-----+		
1. Objectives	1. Objectives	
- Arch. evaluation	- Formal ABI model	
- Security analysis	- Security proofs	
- Interop assessment	- Normative formal analysis	
- Normative mapping	- Compiler & builds theory	
- Cert. roadmap	- Publications & dissertation	
+-----+		
2. Research Questions	2. Research Questions	
- Minimal ABI & audit	- Formal ABI modeling	
- FIPS coverage	- Determinism & safety proofs	
- Interop risks	- FIPS formal verification	
- Reproducible builds	- Interop formal constraints	
+-----+		
3. Domains	3. Domains	
- Crypto & architecture	- Formal verification	
- ABI & interop	- Math modeling	
- Compiler behavior	- Compiler theory	
- Normative standards	- Normative security	
- Threat modeling	- Reproducible systems	
+-----+		
4. Methods	4. Methods	
- ABI inspection	- Coq, Dafny, Lean, TLA+, Z3	
- Interop tests	- Formal semantics	
- Compiler comparison	- Theoretical FIPS/NIST mapping	
- Threat modeling	- Proof construction & checking	
+-----+		
5. Milestones	5. Milestones	
M1 Arch doc	M1 Formal ABI model	
M2 Interop suite	M2 Determinism proven	
M3 Normative analysis	M3 Memory safety proofs	
M4 Security eval	M4 Normative formal mapping	
M5 Cert roadmap	M5 Compiler behavior model	

M6 Final thesis	M6 Publications & thesis	
+-----+-----+		
6. Timeline	6. Timeline	
0-6m: Arch, interop	0-12m: Modeling & initial proofs	
6-12m: Security, comp	12-24m: Extended verification	
12-18m: Cert roadmap	24-36m+: Full proofs, dissertation	
+-----+-----+		
7. Outcomes	7. Outcomes	
- Arch & security eval	- Formal models	
- Normative assessment	- Verified properties	
- Cert roadmap	- Normative proofs	
- Recommendations	- Publications & long-term impact	
+-----+-----+		
8. Vision	8. Vision	
Secure, reproducible	Formally verified, certifiable	
modules for critical	crypto architecture for critical	
sectors & public infra	sectors & global research	
+-----+-----+		